

AIVOA AI-POWERED QMS SYSTEM SPECIFICATION & ARCHITECTURE

TECHNICAL DOCUMENTATION & DEPLOYMENT GUIDE

Project Name	AIVOA Customer Complaint Management System
Industry Domain	Pharmaceutical Manufacturing (API & FDF QMS)
Frontend Stack	React UI + Redux Toolkit + Google Inter Font
Backend Stack	FastAPI (Python) + SQLAlchemy ORM
AI Agent Engine	LangGraph State Graph + Groq LLM (gemma2-9b-it)
Database Engine	PostgreSQL / SQLite
Primary AI Tools	1. Log Complaint Tool 2. Edit Tool 3. Doc Extraction
Bonus AI Features	Root Cause & CAPA Engine, Database Records Modal

DETAILED COMPLAINT / INVESTIGATION SUMMARY:

This document specifies the complete technical architecture and user workflow for the AI-Powered Customer Complaint Management System built for AIVOA Round 1 Full Stack Developer Assessment. The system automates complaint intake, field extraction, context-preserving natural language editing, and risk severity reasoning using LangGraph agent orchestration.